

DESIGN — User-Type Role Model (v2 proposal)

Status: DRAFT / proposal, pending ratification. Not yet a contract. If adopted, this replaces the current two-catalog role setup and updates SPEC tasks roles mrr fees.md §8 + the ARCHITECTURE.md "one task-routing role catalog" invariant. **Authors:** Kevin (concept), Claude (structure/analysis). Started 2026-07-09.

1. Why change

Today OptimumQ has **two overlapping role systems** that both try to gate request work, with nothing keeping them in sync:

- Permission roles** (user_permission_roles → FEE_MANAGER, FEE_AUTHORITY, SEARCH_AND_TRIAGE, REDACTION_WORKER, ...) — used by **task routing** (who a task lands on).
- Function roles** (user_function_roles → SYSTEM_ADMIN, DIRECTOR, SUPERVISOR, ...) — used by **requireRole** middleware (who may act at an endpoint).

Because they overlap and aren't linked, you get **holes**: e.g. the fee-waiver decision endpoint gated on FEE_WAIVER_APPROVER, a role that exists in neither routable catalog — so a task could route to someone who can't act, or an authorized person might never see the task. This violates the ARCHITECTURE.md invariant **"one task-routing role catalog."**

The fix is not "fewer dimensions" — it's dimensions that govern **different, non-overlapping questions**, all hanging off one organizing concept.

2. The organizing concept: User Type

A user is assigned one (or more) **user types** describing the kind of work they do / their position. The user type is the single object management reasons about, and it drives everything below.

3. Three axes (non-overlapping — this is what kills the holes)

Axis	Answers	Set by
Task types	"What request work lands on my list?"	picked per person (a subset of the user type's task menu)
Authority	"What can I do to requests/tasks I don't own?" (reassign, re-route, override, escalate, reopen)	the user type

Axis	Answers	Set by
Permissions	"What system settings/rules can I <i>change</i> ?" (redaction rules, fee profiles, users, taxonomy)	the user type

Each answers a different question, so nothing can fall out of sync. Routing **and** the authority to act both derive from the same assignment — that is the reconciliation.

3a. Two flavors of "task type"

- **Routable work** — estimate, record search, redaction, fee-waiver approval, legal review, MRR tasks. These flow through the workflow engine and land on My-Tasks lists.
- **Non-routable capabilities** — view dashboards, run reports. Never routed; just gate UI. Same picker, different behavior. (Covers City Management, and the reporting side of every type.)

3b. Per-person subsetting

Only the **task-type subset** is chosen per person — because that is the only thing that genuinely varies within a position. Example: a fulfillment team with 12 people, all "Team Staff," some doing only estimates, some only redaction, some all three. Authority and permissions do **not** vary per person; they are properties of the user type.

3c. Multiple user types per person (and the exception mechanism)

A person may hold more than one user type. This handles small cities (one person is Team Supervisor **and** Team Staff doing redaction), and it **is the built-in exception mechanism** for permissions: if a Team Supervisor genuinely needs to edit fee profiles, give them the office user type that carries that permission — no special per-person override machinery in v1.

4. User-type catalog

Office-level types are **team-agnostic** (cross-cutting). Team-level types are **per fulfillment team** (prefixed with the team name). "Authority" and "Permissions" columns are properties of the type; "Task menu" is the set a person of this type can be assigned a subset of.

Office-level

#	User type	Task menu	Authority	Permission groups
1	City Management	(none — never processes requests)	none	Reporting
2	ORO System Administrator	(none / optional)	system (users, teams)	System Administration, Workflow & Taxonomy, Fee Configuration, Reporting — not Legal Rules (technical-only)
3	ORO Director / Manager	(oversight; may act on any)	highest operational — reassign/re-route/override/escalate/reopen across all teams	Workflow & Taxonomy, Fee Configuration, Legal Rules (may) , Reporting, staff/team management
4	ORO Supervisor	(oversight; may act)	operational — reassign/re-route, escalate	Reporting
5	ORO Senior Legal (city attorney)	Legal Review, (Legal Redaction)	legal decisions (AG, exemptions)	Legal Rules (OWNER) , Reporting
6	ORO Legal Associate	Legal Redaction (advanced), Legal Review support	none special	Reporting
7	ORO Associate	Fee-Waiver Approval, MRR (estimate/search/processing coordination)	(MRR request-management)	Reporting

Team-level (per team)

#	User type	Task menu	Authority	Permission groups
8	[Team] Fulfillment Manager	(may do team tasks)	reassign within team; receives coverage-gap emails ; assigns staff task subsets	(team-scoped staff assignment; no global config)
9	[Team] Fulfillment Supervisor	(may do team tasks)	reassign/assist within team	none — explicitly not Legal Rules
10	[Team] Fulfillment Staff	subset of {Estimate Creation, Record Search, Simple Redaction}	none	none

5. Permission groups (coarse, to stay legible — not per-button)

•**System Administration** — users/staff, teams, connectors, security/auth, system config

- **Workflow & Taxonomy** — the rulebook, record types, categories
- **Fee Configuration** — rates, estimate profiles, thresholds
- **Legal Rules** — redaction rules, exemptions/citations
- **Reporting** — dashboards, reports (view)

Decisions locked (2026-07-09):

- **Legal Rules — Senior Legal owns, Director may, Sys Admin technical-only** (not the owner of rule content). So a **Team Supervisor can do a redaction task but cannot change the rules for it** — the core "do the work vs. set the rules" line.
- **Permissions strictly by user type in v1**; exceptions handled by assigning an additional user type (§3c).

6. Coverage-gap safety net

There is no monitoring today for "a request needs a task nobody covers" (e.g. it tries to advance from estimate to record search but no one on the team holds Record Search). Rather than a coverage matrix, **email the team manager** the moment it happens.

Concrete hook: the central `applyStageTransition / spawnForStage` path (built 2026-07-08/09) is exactly the choke point — when it spawns a stage task and `eligibleUsers(team, taskType)` is empty, fire the manager email. A few lines at a spot that already exists.

7. What this implies (scope — not free)

- **Data migration:** map existing function roles + permission roles assignments onto the new user types. The interim fee-waiver routing (FEE AUTHORITY, built 2026-07-09) becomes **ORO Associate + "Fee-Waiver Approval" task type**.
- **UI to build:** an **edit-user** screen (currently there is only add-staff, no edit), and a way to pick a person's task-type subset. Config screens for user types / permission groups.
- **Authority + permissions become type properties** in the data model — replacing the second catalog rather than layering on it.

8. Open questions

1. **Director vs Sys Admin split** — does Director hold staff/team management (part of System Administration), with only connectors/security reserved to Sys Admin?
(Assumed yes above.)
2. **"Finance" vs "ORO Associate"** — the old spec named a **Finance** role for fee-waiver / commercial-rate / objection approvals. This model folds financial approval into **ORO Associate**. Is that right, or does money need its own authority (Director sign-off)?
Note objections.js approvals also reference this.
3. **Who assigns a person's task subset** — Team Manager for their own team, Sys Admin globally, both?
4. **Commercial-rate approval** — same task type as fee waiver, different trigger (purpose='commercial'); confirm it's an ORO Associate task too.
5. **Task-type master list** — this doc lists the ones in play; a complete canonical enumeration (routable + capabilities) is still to be written before build.

9. Next steps

1. Kevin reviews / edits this doc (push back via the repo).
2. Resolve §8 open questions.
3. Write the canonical task-type + permission-group master list.
4. Only then: data-model + migration + UI plan, ratify
into SPEC tasks roles mrr fees.md §8 and ARCHITECTURE.md, and build.

I thought about two related items that I don't want to overlook....the screens currently built to manage data related to City Departments, Fulfillment teams, and Staff Management are not the greatest. So they need to be changed, and Staff Management and Fulfillment Teams UI are directly impacted by this potential modification. A city might have it's own Open Records Department, but in most cases is an entity within the City Clerk's Office or City Records Office. In either case all city departments are listed as City Departments as they would on an org chart, with the system not being relevant to that information. A UI should allow entering city departments. A UI should allow creation of fulfillment teams, and also contain an Open Records office "team" by default. The UI should allow each team to be associated with departments. If Open Records Department exists as a department on the org chart, it would be included in Departments, and The Open Records Office (team) would be associated (by drop down list of Departments) during set up. If no Open Records Department exists, and is a sub-organization in the City Clerk's Office, that office would be selected. Several fulfillment teams might be created that correspond only to what is clearly an associated department, such as Police Records Fulfillment Teams, would be associated with the Police department. An Open Records Fulfillment

team would likely be created, not to be confused with the Open Records Office. The office is management and administration of all of what happens with respect to all fulfillment. The fulfillment teams process requests. A fulfillment team can be associated with multiple departments. For example Open Records Fulfillment team might process requests for City Clerk's Office, Parks and Recreation, etc. The relationship between the department and fulfillment team is included in the taxonomy, helping map a description of a request to the team that would process it. Open Records Office doesn't process requests, but an Open Records fulfillment team does. So Open Records Office should never exist in the Taxonomy. Again, I mention this because staff will be assigned to Open Records office, as well as fulfillment teams, to for associating employees to a team, Open Records office should be considered a team...but not a fulfillment team. The system /UI does not currently provide this distinction (no Open Records Office exists, only a fulfillment team). I visualize a single "parent" screen that is designed to allow easy management of city department/teams/staff, where currently it is multiple screens individually listed on an overloaded panel and no helpful on screen content (a big complaint....on screen content is very helpful in a demo, which is important).

The other item I want to mention is Smart Routing, which at the moment is limited to use within fulfillment teams. The system has a free form text entry box linked to every staff member. When a request is ROUTED to a department, the next step is ASSIGNMENT of a task to eligible staff. There are different methods available to determine assignment, one of which involves AI checking the content of that text (lets assume we're talking about assigning the request's redaction task to one of several of that fulfillment team's members who are eligible for assignment of the task type...AI reads any existing "smart routing" text content for those eligible team members and if the content matches the request description, it's assigned to that person. It might not need to be altered, but I'm trying to visualize where it could be problematic. Of note, at the moment I believe Smart Routing applies only to Record Search and Redaction.

Answers to Section 8 questions:

1. yes
2. Let's add user type Open Records Office Finance designating is as responsible for financial related approvals. Perhaps the catch all of "associate" is vague, but these are not job titles, simply terms used that help make the model intuitive, and as of now the only clearly define tasks type I see that it would include is management of multiple record request if finance related tasks are associated with Open Records Office Finance, but more than those user types would be assigned to different people.
3. System Admin, ORO Manager, both global, Fulfillment team manager for his team
4. Same path and approver for Fee Waiver Request and Commercial rates (new user type will be approver, ORO Finance)
5. This appears to be a comment, so no answer.

I believe that system notifications are sent in a few parts of the application, such as reminders to update legislative content for fees, redaction, timing. I don't recall off the top of my head where those exist other than what I mentioned. I'm not aware of any code for notifications that would make more sense to appear on my tasks lists, but I thought I would mention it just in case some code exists and this structure would cause malfunction.

Regarding existing data migration, if it's problematic, I am ok with manually creating new requests and attempting to process, as most of those were entered to test just the agent itself and the request descriptions would not find results based what documents/records exist.